

PANAMA CITY, FL, USA

WMO: 722245

Lat: 30.212N

Lon: 85.683W

Elev: 6

StdP: 101.25

Time Zone: -6.00 (NAC)

Period: 98-10

WBAN: 03882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	2.1	-9.9	1.6	3.2	-7.5	2.0	5.6	9.8	12.5	8.8	11.6	3.4	340	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	33.8	24.9	32.8	24.9	32.3	24.9	27.4	30.5	26.8	30.2	26.3	29.9	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	28.8	26.0	21.4	28.7	25.2	20.3	28.4	86.4	30.4	83.5	29.8	81.5	30.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.5	6.7	-3.3	36.4	1.9	1.3	-4.6	37.4	-5.7	38.2	-6.7	38.9	-8.1	39.9		
			-5.4	28.5	1.7	1.0	-6.6	29.3	-7.6	29.9	-8.6	30.4	-9.8	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	12.2	13.3	16.7	20.1	24.3	27.3	28.2	28.1	26.4	22.1	17.0	13.2
	DBStd	6.78	5.29	4.12	3.71	3.11	2.26	1.51	1.44	1.47	2.10	3.80	4.07	4.79
	HDD10.0	82	41	14	2	0	0	0	0	0	0	0	3	22
	HDD18.3	690	196	145	74	19	0	0	0	0	0	14	72	171
	CDD10.0	4015	109	108	211	304	442	518	564	560	492	375	212	121
	CDD18.3	1582	6	5	24	73	184	268	306	302	242	131	31	11
	CDH23.3	14702	6	7	74	343	1434	2719	3456	3293	2256	943	152	19
	CDH26.7	5179	0	0	2	36	354	996	1421	1319	802	238	11	0
Wind	WSAvg	3.1	3.3	3.1	3.3	3.3	3.2	2.8	2.6	2.5	3.2	3.0	3.0	3.3
Precipitation	PrecAvg	1352	113	112	131	94	58	118	145	172	142	83	89	94
	PrecMax	1862	439	270	269	252	200	237	254	326	590	222	243	233
	PrecMin	931	36	21	1	20	0	15	46	72	2	15	13	21
	PrecStd	256	88	69	72	68	50	59	57	70	117	54	50	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	24.1	26.7	29.2	32.4	35.0	36.0	34.6	33.9	32.0	27.9	25.2
		MCWB	19.6	18.9	19.4	19.3	21.7	24.6	26.1	25.4	23.2	23.2	21.8	20.4
	2%	DB	22.7	22.7	25.2	27.7	31.0	33.0	33.6	33.1	32.6	30.6	27.0	23.7
		MCWB	19.3	18.3	18.7	20.4	22.0	24.5	25.4	25.7	23.5	23.3	21.4	20.1
	5%	DB	21.8	21.9	23.9	26.9	29.7	32.2	32.7	32.5	32.0	29.0	25.2	22.4
		MCWB	19.2	17.9	18.3	20.2	22.0	24.3	25.5	25.6	23.8	22.7	20.4	19.7
	10%	DB	20.3	20.2	22.7	25.7	28.7	31.1	32.1	32.1	31.0	27.8	23.8	21.2
		MCWB	17.8	17.1	18.1	19.5	22.3	24.1	25.5	25.4	24.0	22.1	19.6	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	21.3	22.5	23.9	25.6	27.2	28.9	28.2	26.8	26.5	24.5	22.3
		MCDB	22.4	22.6	24.3	27.0	28.9	31.3	31.4	31.2	29.5	28.6	26.7	23.8
	2%	WB	20.5	20.2	21.3	23.0	24.8	26.4	27.7	27.4	26.0	25.4	23.3	21.3
		MCDB	21.9	21.6	23.0	25.8	28.2	30.3	30.8	30.2	29.2	28.2	25.2	22.6
	5%	WB	19.4	19.2	20.4	22.3	24.3	25.8	26.9	26.8	25.6	24.4	22.2	20.3
		MCDB	21.0	20.5	22.5	24.7	27.6	29.6	30.5	30.1	29.1	27.5	24.1	21.7
	10%	WB	18.3	17.9	19.2	21.4	23.7	25.4	26.2	26.3	25.2	23.7	20.8	19.1
		MCDB	19.8	19.1	21.8	23.7	26.9	29.2	30.1	29.9	28.9	26.6	23.2	20.8
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.1	10.0	9.5	8.9	7.9	7.6	7.6	8.4	10.0	10.7	10.0
		MCDBR	9.0	10.1	10.1	10.0	9.8	9.0	8.5	8.4	9.4	9.8	10.6	9.4
		MCWBR	6.7	6.8	5.9	4.7	3.9	3.4	3.5	3.4	3.6	4.2	5.9	6.9
	5% WB	MCDBR	8.3	8.3	8.1	7.4	6.9	7.7	7.7	7.5	7.3	7.9	7.9	8.7
		MCWBR	7.2	7.2	6.0	4.5	3.6	3.3	4.0	3.6	3.4	4.4	5.6	7.6
Clear-Sky Solar Irradiance	taub	0.335	0.349	0.368	0.395	0.437	0.468	0.498	0.493	0.436	0.384	0.355	0.343	
	taud	2.536	2.499	2.450	2.375	2.290	2.260	2.191	2.202	2.338	2.465	2.482	2.526	
	Ebn at Noon	899	916	918	899	858	827	800	801	838	866	866	871	
	Edh at Noon	87	98	110	122	134	138	147	144	121	100	91	84	
All-Sky Solar Radiation	RadAvg	3.13	3.78	5.04	6.06	6.67	6.18	5.83	5.43	5.06	4.52	3.53	2.78	
	RadStd	0.23	0.47	0.39	0.49	0.42	0.47	0.38	0.40	0.33	0.40	0.29	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-100	+167	+104

Nomenclature: See separate page